

CHANDIGARH UNIVERSITY APEX INSTITUTE OF TECHNOLOGY



ASSIGNMENT 1

Subject: Computer Networks
Subject Code: 24CST-339

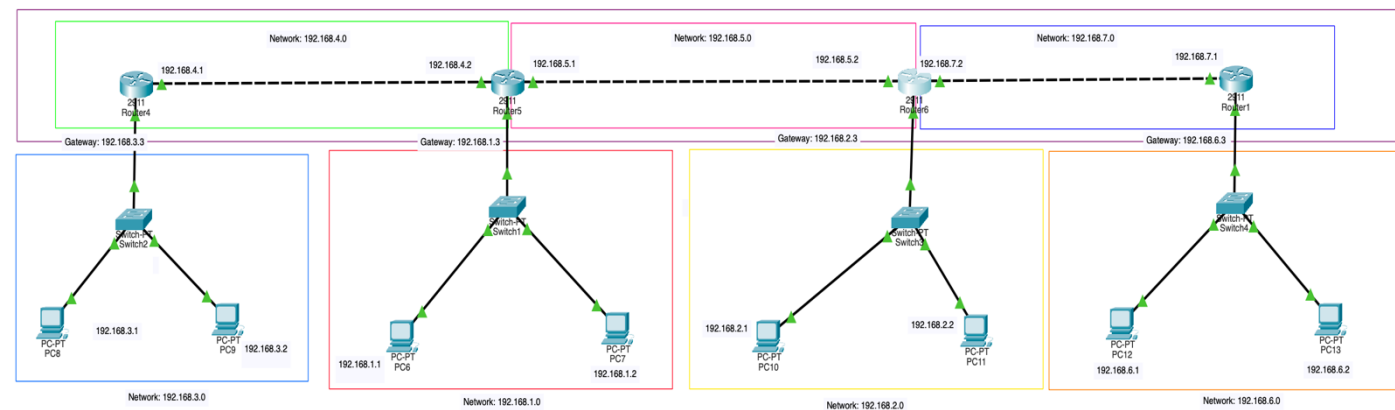
Submitted By:
Name: Vishesh Kansugia
UID: 24BAI70184
Class: 24AIT_KRG_1B

Submitted To:
Mr. Alok Singh Rathore

Question 1.

Develop Static Routing with 4 routers, 4 Switches & 8 Pc's. Make sure to highlight the Networks with their network ID's. IP Addresses should be written in text format as well.

Answer:



Static Routing Table:

Router	Destination Network	Subnet Mask	Next Hop
Router4	192.168.1.0	255.255.255.0	192.168.4.2
	192.168.2.0	255.255.255.0	192.168.4.2
	192.168.6.0	255.255.255.0	192.168.4.2
Router5	192.168.3.0	255.255.255.0	192.168.4.1
	192.168.2.0	255.255.255.0	192.168.5.2
	192.168.6.0	255.255.255.0	192.168.5.2
Router6	192.168.1.0	255.255.255.0	192.168.5.1
	192.168.3.0	255.255.255.0	192.168.5.1
	192.168.6.0	255.255.255.0	192.168.7.1
Router1	192.168.2.0	255.255.255.0	192.168.7.2
	192.168.1.0	255.255.255.0	192.168.7.2
	192.168.3.0	255.255.255.0	192.168.7.2

Network Table:

Network ID	Purpose	Devices Connected
192.168.1.0	LAN	Router5, PC6, PC7
192.168.2.0	LAN	Router6, PC10, PC11
192.168.3.0	LAN	Router4, PC8, PC9
192.168.4.0	WAN	Router4 to Router5
192.168.5.0	WAN	Router5 to Router6
192.168.6.0	LAN	Router1, PC12, PC13
192.168.7.0	WAN	Router6 to Router1

Question 2:

Write the difference between Hub, Switch & Router in Tabular Format. Explain why Router is the most intelligent and Hub is the least intelligent device.

Answer:

Difference between Hub, Switch and Router

Feature	Hub	Switch	Router
OSI Layer	Layer 1 (Physical)	Layer 2 (Data Link)	Layer 3 (Network)
Address Used	No Address	MAC Address	IP Address
Data Forwarding	Broadcasts data to all devices	Sends data only to the destination device	Routes data between different networks
Decision Making	No	Yes (using MAC Address Table)	Yes (using Routing Table)
Collision Domain	One	One per port	One per interface
Broadcast Domain	One	One (without VLANs)	Multiple
Speed	Slow	Faster	Fastest
Security	Low	Moderate	High
Main Use	Connects devices in a LAN	Connects devices efficiently within a LAN	Connects different networks

Why is Router the Most Intelligent Device?

A router is considered the most intelligent networking device because:

- It works at the Network Layer (Layer 3).
- It uses IP addresses to identify the destination network.
- It maintains a routing table to determine the best path for data transmission.
- It connects different networks such as LANs and WANs.
- It reduces unnecessary traffic by forwarding packets only to the correct network.

Why is Hub the Least Intelligent Device?

A hub is considered the least intelligent networking device because:

- It works at the Physical Layer (Layer 1).
- It does not use MAC or IP addresses.
- It simply broadcasts data to all connected devices, regardless of the destination.
- It cannot make forwarding decisions.
- It creates unnecessary network traffic and increases collisions.

A Hub is a basic device that broadcasts data to all devices, a Switch forwards data using MAC addresses within a LAN, and a Router connects different networks using IP addresses and selects the best path for data transmission. Therefore, the router is the most intelligent, while the hub is the least intelligent networking device.

Question 3:

What are Random Access Protocols? Why do we need them? Explain each category with diagram and example. Also explain the difference between CSMA/CD and CSMA/CA.

Answer:

In a computer network, many devices often share the same communication channel. If two or more devices transmit data at the same time, their data packets collide, resulting in data loss. To avoid or reduce such collisions, Random Access Protocols are used.

A Random Access Protocol is a Medium Access Control (MAC) protocol that allows any device to transmit data whenever it has data to send. Before transmitting, the device checks whether the communication channel is free. If the channel is busy or a collision occurs, the protocol follows specific rules to retransmit the data.

Need for Random Access Protocols

Random Access Protocols are required because they:

- Allow multiple devices to share the same communication channel.
- Reduce data collisions.
- Improve network efficiency.
- Ensure fair access to the communication medium.
- Prevent unnecessary retransmissions and network congestion.

Categories of Random Access Protocols

The main categories of Random Access Protocols are:

1. ALOHA
2. CSMA (Carrier Sense Multiple Access)
3. CSMA/CD
4. CSMA/CA

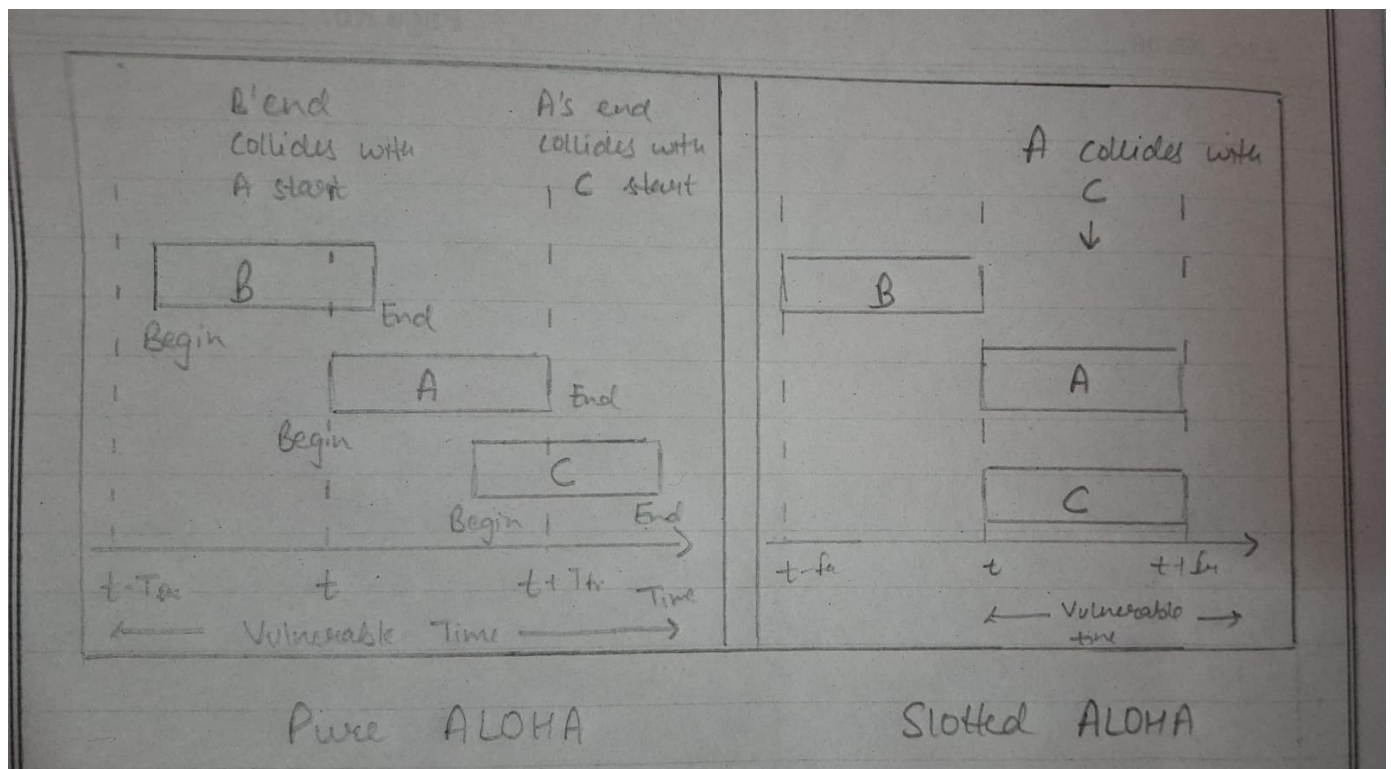
1. ALOHA

ALOHA is the simplest random access protocol. In this protocol, a device transmits data whenever it has data to send without checking whether the communication channel is busy or free. If two devices transmit at the same time, a collision occurs. The sender waits for a random amount of time and retransmits the data.

Working

- A station sends data immediately.
- If an acknowledgement (ACK) is received, transmission is successful.
- If no ACK is received, the sender assumes a collision has occurred.
- The sender waits for a random time and retransmits the data.

Diagram



Example

Suppose Computer A and Computer B both send data at the same time.

Since both transmissions overlap, a collision occurs. Both computers wait for a random time and then retransmit their data.

Types of ALOHA

- Pure ALOHA
- Slotted ALOHA

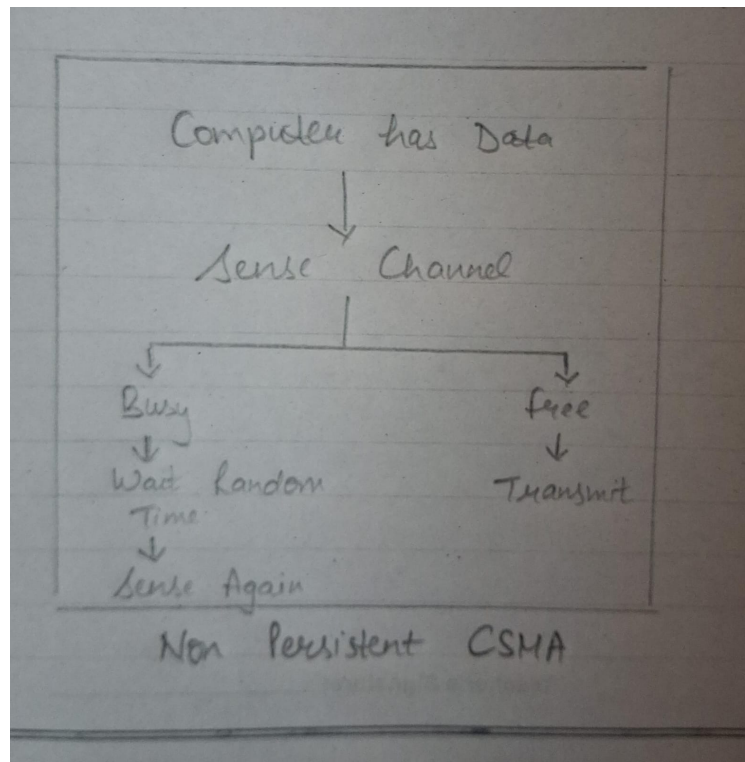
2. CSMA (Carrier Sense Multiple Access)

CSMA improves upon ALOHA by checking the communication channel before transmitting data. A device first listens to the channel. If the channel is idle, it transmits the data. If the channel is busy, it waits until the channel becomes free.

Working

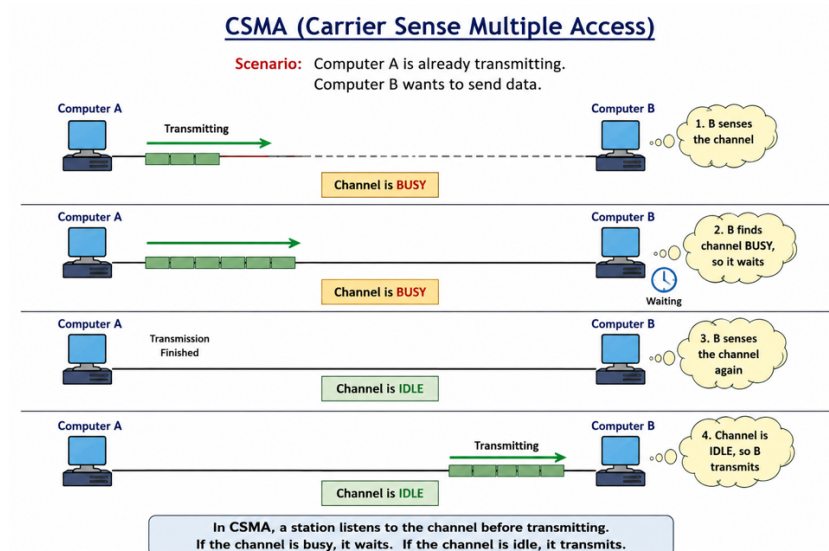
- Listen to the channel.
- If the channel is idle, transmit data.
- If the channel is busy, wait.
- Try transmitting again after the channel becomes free.

Diagram



Example

Suppose Computer A is already transmitting.
 Computer B wants to send data.
 Instead of transmitting immediately, Computer B first checks the channel.
 Since the channel is busy, it waits until Computer A finishes transmitting.
 Thus, collisions are reduced.



3. CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

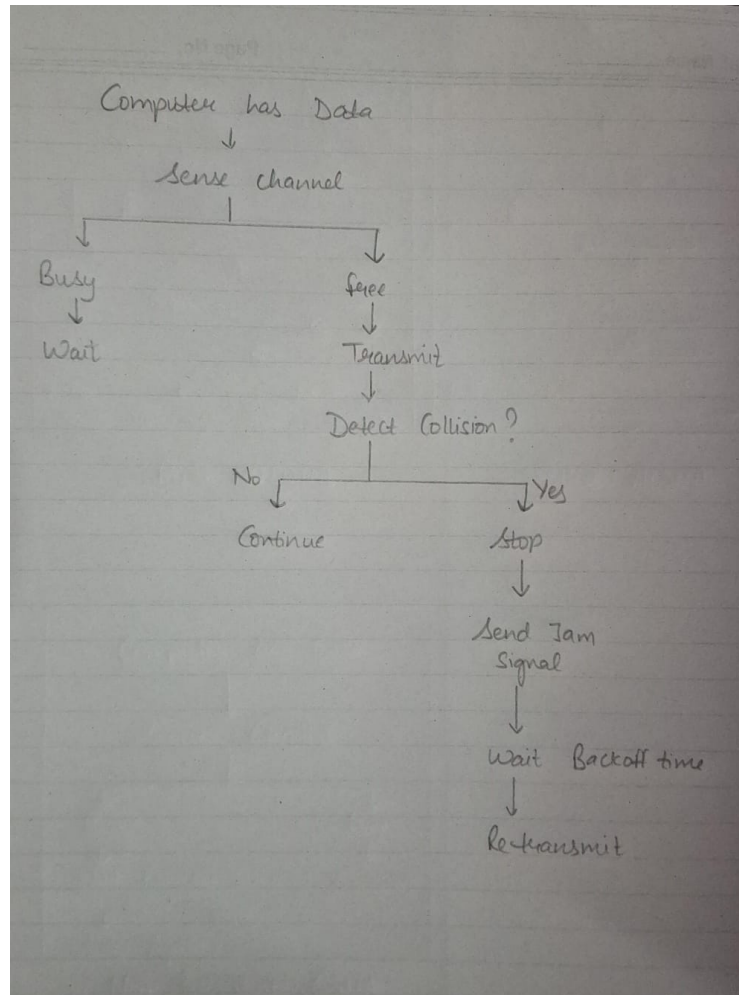
CSMA/CD is an improved version of CSMA used in wired Ethernet networks. In addition to checking whether the channel is free, the sender also monitors the channel while transmitting. If a collision occurs, it immediately stops transmission and sends a Jam Signal so that all devices know a collision has occurred.

Working

1. Listen to the channel.

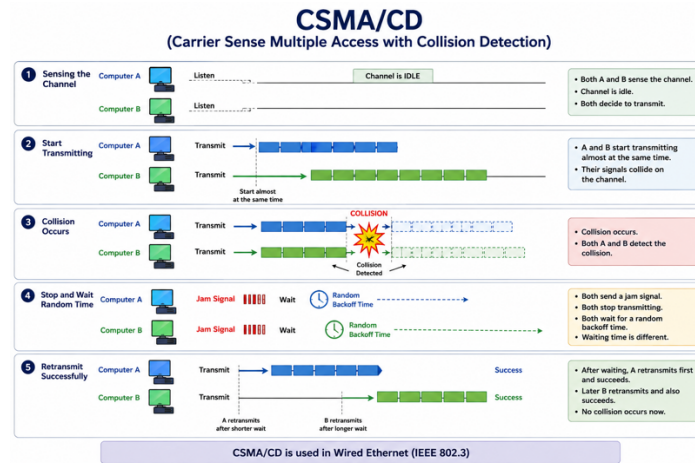
2. If the channel is free, start transmitting.
3. Continue monitoring while transmitting.
4. If a collision occurs:
 - Stop transmission.
 - Send a Jam Signal.
 - Wait for a random backoff time.
 - Retransmit the data.

Diagram



Example

Computer A and Computer B start transmitting almost simultaneously.
Both detect the collision.
Both stop transmitting immediately.
After waiting for different random times, they retransmit the data.



Applications

- Wired Ethernet (IEEE 802.3)

4. CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

CSMA/CA is used in wireless networks (Wi-Fi). Since wireless devices cannot detect collisions while transmitting, they try to avoid collisions before transmission.

Before sending data, the sender waits for the channel to become idle. It may also exchange RTS (Request to Send) and CTS (Clear to Send) messages before transmitting.

Working

1. Listen to the channel.
2. If busy, wait.
3. If free, send RTS.
4. Receiver replies with CTS.
5. Sender transmits data.
6. Receiver sends ACK.

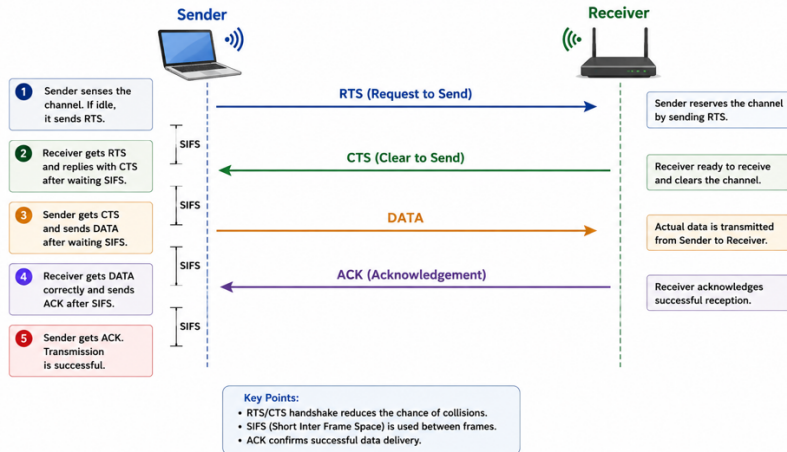
Example

Suppose a laptop wants to send data to a Wi-Fi router.
 The laptop first sends an RTS message.
 The router replies with CTS.
 Only after receiving CTS does the laptop transmit the data.
 This minimizes the chances of collisions.

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

Used in Wireless Networks (IEEE 802.11 Wi-Fi)

Before sending data, the sender checks the channel and uses RTS/CTS handshake to avoid collisions.



Applications

- Wi-Fi (IEEE 802.11)

Difference between CSMA/CD and CSMA/CA

Basis	CSMA/CD	CSMA/CA
Full Form	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Working Principle	Detects collisions after transmission starts	Avoids collisions before transmission
Network Type	Wired Networks	Wireless Networks
Standard	IEEE 802.3	IEEE 802.11
Collision Handling	Detects and retransmits after collision	Prevents collisions using RTS/CTS
Jam Signal	Used	Not Used
RTS/CTS	Not Used	Used
Efficiency	High in wired LAN	High in wireless LAN
Example	Ethernet	Wi-Fi